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The EcoCyc and MetaCyc Databases

Instrumental Reading Report

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1. Introduction

This report was developed from the analysis of the scientific article “*The EcoCyc and MetaCyc Databases*”, published in the journal *Nucleic Acids Research* in 2000.

The objective of this activity was to identify technical English linguistic structures such as cognates, false cognates, noun phrases, modal verbs, linking words, and affixes, as well as to apply reading techniques like *Skimming* and *Scanning*.

The chosen article relates to the areas of Bioinformatics, Databases, and Scientific Computing.

2. Legend

The following color codes were applied directly to the PDF file of the article.

Color	Description
	Cognates – Words that are similar in form and meaning to Portuguese (e.g., database, metabolic, enzyme)
	False Cognates – Words that look similar but have different meanings
	Noun Phrases – Groups of words centered around a noun (e.g., <i>complete inventory of compounds</i>)
	Verb To Be – am, is, are, was, were, be, being, been
	Verb Tenses – Markers indicating time of action
	Modal Verbs – can, could, may, might, must, shall, should, will, would
	Prefixes – Added to the beginning of a word (e.g., re-, inter-, sub-)
	Suffixes – Added to the end of a word (e.g., -tion, -ment, -ive)
	Linking Words – however, therefore, because, although, and, but

These colors were used to highlight the corresponding linguistic elements throughout the PDF article.

3. Skimming — General Idea

The article describes two bioinformatics databases: EcoCyc and MetaCyc.

EcoCyc is an organism-specific database for *Escherichia coli*, describing metabolic

pathways, transporters, genes, and cellular processes.

MetaCyc is a broader metabolic pathway database containing information from various organisms, mainly microorganisms.

Both databases use the software *Pathway Tools*, which allows queries, visualizations, and organization of biological data using computational concepts such as databases, object-oriented modeling, and knowledge representation systems.

The article also presents improvements in version 5.0 of both systems, especially the inclusion of membrane transport systems and new query tools.

4. Scanning — Specific Data Location

Table 1 summarizes specific data found through scanning.

Tabela 1: Specific data located in the article

Information sought	Result found
EcoCyc 5.0 release date	June 1999
MetaCyc 5.0 release date	September 1999
Query software name	Pathway Tools
Knowledge representation system	Ocelot (FRS)
Number of transporters imported into EcoCyc	Over 300
Number of Enzyme II complexes described in v5.0	20
Percentage of ORFs with no attributed function	32%
Percentage of genes with putative function	21%
Number of genes in EcoCyc	4390
Number of compounds in EcoCyc	1868
Number of compounds in MetaCyc	1963
Research funding	Grant 1-R01-RR07861-01

Information sought	Result found
Main access URL	http://ecocyc.PangeaSystems.com/ecocyc/

5. Glossary of Computing Terms

Table 2 presents the main computing terms found in the article.

Tabela 2: Glossary of computing terms

Term	Translation	Concept
Database	Banco de dados	System for storing and organizing information
Schema	Esquema	Logical structure of data
Query	Consulta	Request for data from a database
GUI	Interface gráfica	Visual interface for interaction
HTML Form	Formulário HTML	Web structure for data submission
Object-oriented data model	Modelo orientado a objetos	Data organization as objects
Dataset	Conjunto de dados	Organized collection of information
Visualization tools	Ferramentas de visualização	Graphical resources for data analysis
Structured query	Consulta estruturada	Search with defined parameters
Unique identifier	Identificador único	Exclusive identification code
Constraint	Restrição	Condition applied to data
Attributes	Atributos	Characteristics of an object
Classes	Classes	Grouping of similar objects

Term	Translation	Concept
License	Licença	Legal permission for software use

6. Complex Sentence Analysis

Three complex sentences from the article are analyzed below using instrumental reading techniques.

6.1. Sentence 1

“The EcoCyc and MetaCyc databases are online reference sources for metabolic data.”

- **Main noun phrase:** The EcoCyc and MetaCyc databases
- **Verb To Be:** are
- **Verb tense:** Simple Present
- **Cognates:** databases, reference, metabolic
- **Structure:** Compound subject + linking verb + predicate

6.2. Sentence 2

“MetaCyc is a new metabolic-pathway database that describes pathways and enzymes of many different organisms, with a microbial focus.”

- **Noun phrase:** a new metabolic-pathway database
- **Verb To Be:** is
- **Main verb:** describes
- **Prefix:** meta-
- **Cognates:** database, metabolic, pathways, enzymes, organisms
- **Linking word:** that

6.3. Sentence 3

“We seek to make the annotation of the E. coli genome within EcoCyc as complete and up-to-date as possible based on new functional characterizations.”

- **Verb tense:** Simple Present
- **Expression:** *as ... as possible* indicates maximum completeness
- **Prefixes:** **an-** (annotation), **func-** (functional)
- **Suffix:** **-tion** (characterizations)
- **Long noun phrase:** the annotation of the E. coli genome within EcoCyc
- **Cognates:** annotation, genome, functional

7. Research Source

The article was found using the platform *Research Rabbit*, an AI-based tool for discovering and recommending scientific articles.

The choice of this article was due to:

- The platform’s ease of navigation.
- Immediate availability of the PDF file.
- The topic’s relation to computing areas such as Databases, Bioinformatics, and Information Systems.
- The presence of numerous technical terms in English, contributing to technical vocabulary learning.

8. Conclusion

This activity helped develop technical English reading and interpretation skills through the use of *Skimming* and *Scanning* techniques.

It was also possible to identify several computational concepts applied to Bioinformatics, especially involving databases, data modeling, and information systems.

The work contributed both to technical vocabulary improvement and to the understanding of computing applications in scientific research.

9. References

[1] KARP, P. D.; RILEY, M.; SAIER, M.; PAULSEN, I. T.; PALEY, S. M.; PELLEGRINI-TOOLE, A. *The EcoCyc and MetaCyc Databases*. **Nucleic Acids Research**, v. 28, n. 1, p. 56–59, 2000.

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